

How Much Infinity Does Physics Need? Reverse Mathematics of Physical Limits and the Empirical Boundary

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Abstract

Infinite limits are everywhere in physics: thermodynamic limits, infrared limits of renormalisation-group flows, fractal dimensions, and limit sets. We ask how much set existence such limits require, in the sense of reverse mathematics, and we compare the answer with what finite experiments can certify.

We prove four equivalences over RCA_0 . Each of the following is equivalent to arithmetical comprehension ACA_0 : Fekete’s lemma for subadditive sequences, which underlies the existence of thermodynamic limits; the existence of the infrared limit of every bounded monotone c -function; the existence of the scaling exponent of every Cantor construction with nondecreasing dimension parameters; and the existence of the set of indices whose computable fractal limit sets are connected. At the level of individual quantities, a limit whose convergence has a computable modulus that RCA_0 can verify is provably existent in RCA_0 . A limit that is not a computable real is not, because the ω -model REC of computable sets contains no real equal to it.

We classify the limits used in a series of nine papers accordingly. The quantities for which explicit certification budgets were found come with explicit moduli and are provable in RCA_0 . This includes the Ising free energy, whose finite-size error is $0.639913/L^2$ at criticality, matching the Ising CFT torus amplitude $\ln[(\theta_2 + \theta_3 + \theta_4)/2\eta](i) = 0.639912$. It also includes the critical-cluster mass of the diamond lattice, with geometric modulus $\Lambda_2/\Lambda_1 = 0.3553$, and the embezzlement asymptotics, with error $O(1/(n \ln n))$. The quantities that need ACA_0 are exactly those defined by limits without computable modulus: a fractal dimension equal to a function of Chaitin’s Ω , the connectivity of limit sets, and the finite-dimensional values of nonlocal games. These are precisely the quantities shown elsewhere to be empirically inaccessible. Since ACA_0 is conservative over Peano arithmetic, the infinity that these physical limits require is real but logically mild. The only result in the series whose logical strength is unknown is the classification of injective von Neumann algebras.

Contents

1	Introduction	2
2	Background	2
3	Four equivalences	2
3.1	Thermodynamic limits: Fekete’s lemma	2
3.2	Infrared limits of c -functions	4
3.3	Fractal scaling exponents	4
3.4	Connectivity of computable limit sets	5

4	Individual limits: computable moduli	5
5	The empirical boundary	6
6	Discussion	6

1 Introduction

Reverse mathematics asks which set-existence axioms are needed to prove a theorem of ordinary mathematics [1, 2]. Most theorems of analysis turn out to be provable in, or equivalent over a weak base theory to, one of a few subsystems of second-order arithmetic. Feferman argued that the mathematics actually used in the sciences can be formalised in a system conservative over Peano arithmetic [3]. Physics thus needs only a little of the infinite.

This paper makes that statement precise for the limits that occur in physics, and relates it to a different question: which limits can be certified by finite experiments? In a series of papers we found sharp answers to the second question. Fractal dimensions can be uncomputable [4]. Connectivity of computable limit sets is undecidable, with no computable critical scale [5]. Certifying Hilbert-space dimension beyond holographic bounds exceeds available resources [6], and relativistic quantum field theory closes the one loophole [7]. Elsewhere, explicit finite-size bounds were found: for the diamond-lattice percolation cluster [8], for embezzlement [6], and for the effective dimension of the vacuum [7].

We show that the two classifications coincide. The limits with explicit certification budgets have explicit moduli of convergence and are provable in RCA_0 . The limits shown to be empirically inaccessible require arithmetical comprehension, and their uniform versions are equivalent to ACA_0 . Figure 1 illustrates the distinction numerically.

2 Background

We use the standard subsystems of second-order arithmetic [2]. RCA_0 consists of basic arithmetic, Σ_1^0 -induction and Δ_1^0 -comprehension; its sets are, informally, those computable from given sets. ACA_0 adds comprehension for arithmetical formulas. The collection REC of all computable sets is an ω -model of RCA_0 but not of ACA_0 . A real number is coded by a sequence of rationals (q_k) with $|q_k - q_l| \leq 2^{-k}$ for $l > k$. In REC the reals are exactly the computable reals. ACA_0 is conservative over PA for arithmetical sentences [2].

We use two facts [2, §§III.1–III.2].

(F1) Over RCA_0 , ACA_0 is equivalent to: every bounded nondecreasing sequence of rational numbers converges.

(F2) Over RCA_0 , ACA_0 is equivalent to: for every set X , the Turing jump X' exists.

The reversal in (F1) goes back to Specker’s computable bounded increasing sequence of rationals with a non-computable limit [9].

3 Four equivalences

3.1 Thermodynamic limits: Fekete’s lemma

The existence of the free energy per site in the thermodynamic limit rests, for translation-invariant interactions, on subadditivity of $-\ln Z$ and on Fekete’s lemma [10, 11].

Theorem 1. *Over RCA_0 the following are equivalent:*

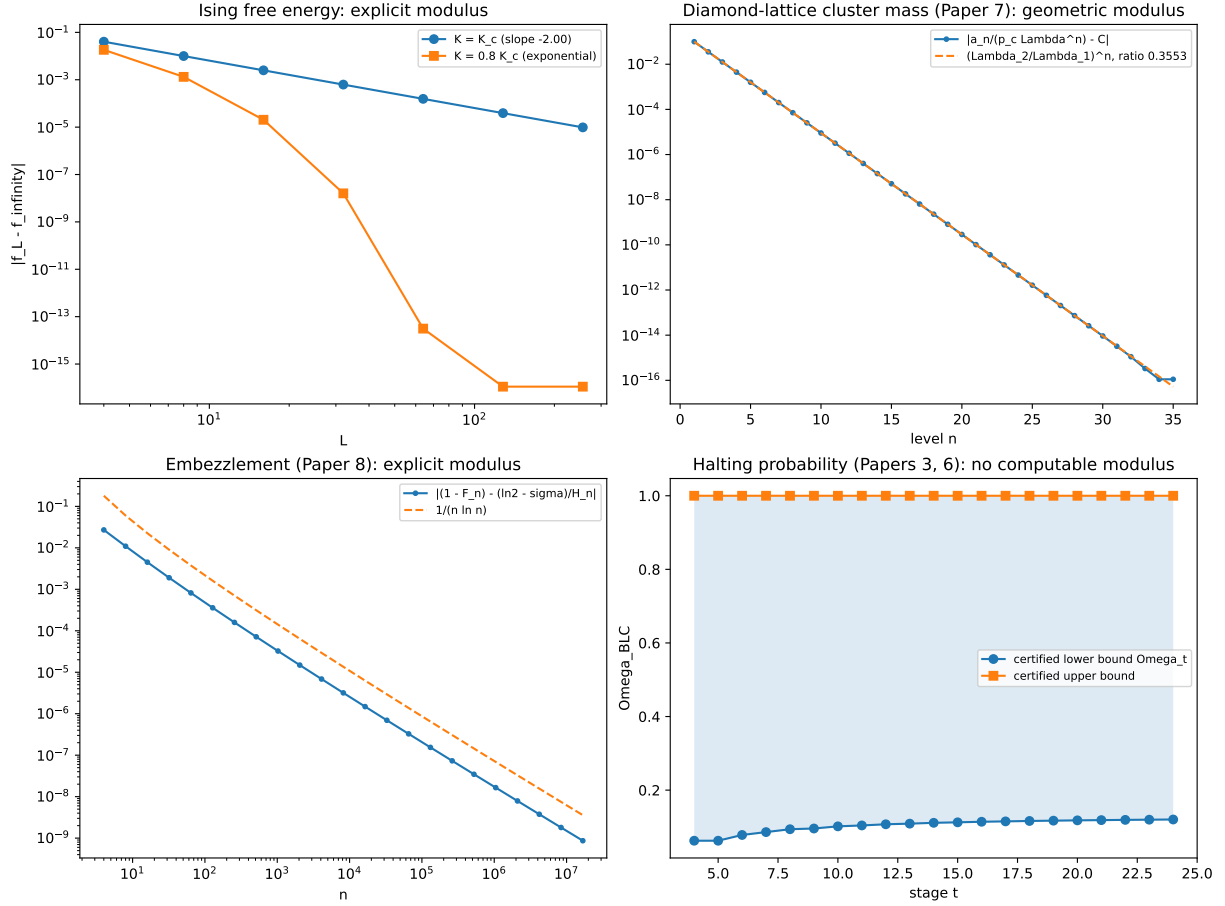


Figure 1: Limits with and without computable moduli. *Top left:* error of the exact free energy per site of the two-dimensional Ising model on an $L \times L$ torus, at K_c ($0.639913/L^2$) and at $0.8K_c$ (exponential). *Top right:* convergence of the critical-cluster mass of the diamond lattice, with ratio $\Lambda_2/\Lambda_1 = 0.3553$. *Bottom left:* error of the embezzlement asymptotics, below $1/(n \ln n)$. *Bottom right:* certified lower and upper bounds for the halting probability of binary lambda calculus. The lower bound increases, but no computable bound on the remaining distance exists, and the certified interval does not shrink.

(i) ACA_0 ;

(ii) for every sequence of reals (a_n) with $a_n \geq 0$ and $a_{m+n} \leq a_m + a_n$, the limit $\lim_n a_n/n$ exists.

Proof. (i) $\Rightarrow$ (ii). In ACA_0 the infimum $\lambda = \inf_n a_n/n$ exists, by the least-upper-bound principle for sequences, which is equivalent to (F1) [2]. The classical argument then applies. Given m , write $n = qm + r$ with $0 \leq r < m$, and set $a_0 = 0$. Then $a_n \leq qa_m + a_r$, so $a_n/n \leq a_m/m + \max_{r < m} a_r/n$, which tends to a_m/m . This uses only Σ_1^0 -induction. Hence $\lim a_n/n = \lambda$.

(ii) $\Rightarrow$ (i). Let (c_n) be a nondecreasing sequence of rationals in $[0, 1]$, and put $b_n = 1 - c_n$ and $a_n = nb_n$. Then $a_n \geq 0$, and since b is nonincreasing, $a_{m+n} = (m+n)b_{m+n} \leq mb_m + nb_n = a_m + a_n$. By (ii), $a_n/n = b_n$ converges, hence so does c_n , and (F1) gives ACA_0 . $\square$

Remark. The reversal uses only rational sequences, so the equivalence also holds if (ii) is restricted to rational sequences. Nonnegativity is no restriction: if only $a_n \geq -Cn$, then $\tilde{a}_n = a_n + Cn$ is subadditive and nonnegative, and $\tilde{a}_n/n$ converges if and only if a_n/n does. Thermodynamic limits in d dimensions are proved by the same scheme, first along cubes and then along van Hove sequences, with surface corrections to subadditivity [11]. The argument

uses the existence of an infimum together with arithmetical estimates, so we expect it to go through in ACA_0 . The reversal applies to it a fortiori, but we do not formalise the d -dimensional case here.

Remark. In computability-theoretic terms, Boche, Böck and Deppe showed that the Fekete limit of a computable subadditive sequence is in general not computable, and they classified the associated sets in the arithmetical hierarchy [12]. Theorem 1 is the corresponding statement in reverse mathematics. For a specific model the situation can be better. Onsager’s solution gives the free energy of the two-dimensional Ising model as a computable real with explicit finite-size corrections (Section 4).

3.2 Infrared limits of c -functions

The c -theorem asserts that a suitable c -function decreases along the renormalisation-group flow, and its infrared value is the limit [13]. The following is a restatement of (F1) in this language.

Corollary 3.1. *Over RCA_0 , ACA_0 is equivalent to: for every bounded nonincreasing sequence $c(r_1) \geq c(r_2) \geq \dots$ of rationals, indexed by an increasing sequence of scales, the infrared limit $\lim_k c(r_k)$ exists.*

For a given flow with an explicit rate of approach to its fixed point, the infrared limit exists already in RCA_0 . The uniform statement, “every c -function has an infrared value”, has the strength of ACA_0 .

3.3 Fractal scaling exponents

In [4] we considered homogeneous Cantor sets in which every interval at stage k is replaced by its two end pieces of ratio $r_k = 2^{-1/d_k}$, with $d_k = \frac{1}{4} + \frac{1}{2}a_k$. The box-counting and Hausdorff dimensions equal the scaling exponent

$$D = \lim_{k \rightarrow \infty} \frac{k}{\sum_{j=1}^k 1/d_j} \quad (1)$$

whenever it exists. When a_k are lower approximations of Chaitin’s Ω , $D = \frac{1}{4} + \frac{1}{2}\Omega$ is not computable.

Theorem 2. *Over RCA_0 the following are equivalent:*

- (i) ACA_0 ;
- (ii) *for every nondecreasing sequence (a_k) of rationals in $[0, 1)$ the scaling exponent (1) exists.*

Proof. Put $b_k = 1/d_k \in (4/3, 4]$, a nonincreasing sequence, and let $\bar{b}_k = \frac{1}{k} \sum_{j=1}^k b_j$ be its Cesàro means. Then (1) states that $1/\bar{b}_k$ converges.

(i) $\Rightarrow$ (ii). By (F1) b_k converges to some L . Cesàro means of a convergent sequence converge to the same limit, by an argument that uses only Σ_1^0 -induction, so $D = 1/L$.

(ii) $\Rightarrow$ (i). Let $\bar{b}_k \rightarrow L$. Since b is nonincreasing, $b_k \leq \bar{b}_k$. For $m > k$,

$$\bar{b}_m \leq \frac{k\bar{b}_k + (m-k)b_k}{m} = b_k + \frac{k}{m}(\bar{b}_k - b_k).$$

Suppose $b_k < L$ for some k , and let $\delta = L - b_k > 0$. Choose $m > 2k(\bar{b}_k - b_k)/\delta$ so large that also $\bar{b}_m > L - \delta/2$. Then $\bar{b}_m \leq b_k + \delta/2 = L - \delta/2$, a contradiction. Hence $L \leq b_k$ for all k . This argument involves only finitely many comparisons of given reals and is carried out in RCA_0 . Together with $b_k \leq \bar{b}_k$ it gives $L \leq b_k \leq \bar{b}_k \rightarrow L$, so $b_k \rightarrow L$. Then $a_k = 2(1/b_k - \frac{1}{4})$ converges, and (F1) gives ACA_0 . $\square$

3.4 Connectivity of computable limit sets

In [5] we constructed computable compact sets $K_e \subset [0, 1]^2$ with finite stages $K_{e,n}$, such that the bridge of K_e is cut at stage s exactly when the e -th machine halts at step s . Thus K_e is connected if and only if $K_{e,n}$ is connected for all n , if and only if the machine never halts. The construction relativises to an oracle X .

Theorem 3. *Over RCA_0 the following are equivalent:*

(i) ACA_0 ;

(ii) *for every set X , the set $C^X = \{e : K_{e,n}^X \text{ is connected for every } n\}$ exists.*

Proof. Each $K_{e,n}^X$ is a finite union of dyadic squares, computable from X . By inspection of the construction, RCA_0 proves that $K_{e,n}^X$ is disconnected if and only if $n \geq 5$ and $\Phi_e^X(e)$ halts within n steps. Hence $e \in C^X$ if and only if $\Phi_e^X(e)$ never halts, so C^X is the complement of X' . Thus C^X exists if and only if X' exists, and (F2) gives the equivalence. $\square$

Remark. We formulated connectivity stage by stage. This avoids formalising the topology of compact sets in second-order arithmetic, and it is equivalent to connectivity of the limit set by [5]. The same argument applies to the Ω -comb of [5]. Its attachment pattern computes Ω , and so it does not exist in REC .

4 Individual limits: computable moduli

Equivalences such as Theorems 1–3 concern uniform statements. For a single physical quantity the relevant question is whether its defining limit has a computable modulus of convergence.

Proposition 4.1. *Let (x_n) be a computable sequence of rationals.*

- (a) *If there is a computable f such that RCA_0 proves $|x_m - x_n| \leq 2^{-k}$ for all k and all $m, n \geq f(k)$, then RCA_0 proves that $\lim_n x_n$ exists.*
- (b) *If (x_n) converges to a real that is not computable, then RCA_0 does not prove that $\lim_n x_n$ exists.*

Proof. (a) The sequence $(x_{f(k)})_k$ is a code for a real, and RCA_0 proves that x_n converges to it. (b) REC is an ω -model of RCA_0 . In REC every real is computable, and the statement “ $\lim_n x_n$ exists” would assert that a computable real equals the limit, which is false. By soundness, RCA_0 does not prove the statement. $\square$

We evaluate the moduli of the limits used in the series with the script accompanying this paper.

- *Ising free energy.* From Kaufman’s exact formula on the $L \times L$ torus [14] and Onsager’s solution [15], the error at K_c is $|f_L - f_\infty| = 0.639913/L^2$ at $L = 256$. The constant agrees with the universal torus amplitude of the Ising conformal field theory, $\ln[(\theta_2 + \theta_3 + \theta_4)/(2\eta)]$ at $\tau = i$, equal to 0.639912 [16]. At $K = 0.8K_c$ the error decays as $e^{-0.490L}$. Both are explicit moduli.
- *Critical-cluster mass* [8]. The mass recursion is linear with matrix entries in $\mathbb{Q}(\sqrt{5})$. The normalised mass $a_n/(p_c \Lambda^n)$ converges geometrically with ratio $\Lambda_2/\Lambda_1 = 0.355318$, observed to 10^{-16} .
- *Embezzlement* [6]. The difference between $1 - F_n$ and $(\ln 2 - \sigma)/H_n$ falls from 2.7×10^{-2} at $n = 4$ to 8.7×10^{-10} at $n = 2^{24}$, and $n \ln n$ times it stays below 0.242.
- *Halting probability* [4, 5]. For binary lambda calculus, the certified lower bound rises to 0.1202 at stage 24, while the certified upper bound stays at 1. No computable modulus exists for the halting probability of a universal machine.

5 The empirical boundary

Table 1 collects the limits used in the series, their moduli, and the weakest of the systems considered here that proves their existence.

Table 1: Logical strength of the limits used in the series and their empirical status.

Limit	Modulus	System	Empirical status
2D Ising free energy, finite-size error	$0.64/L^2$ at K_c , exponential off K_c	RCA_0	resolved below experimental precision [6]
Diamond-lattice cluster mass $d_f = \log_2 \Lambda$	geometric, 0.3553^n	RCA_0	computed exactly [8]
Embezzlement fidelity	$O(1/(n \ln n))$	RCA_0	enters explicit certification budget [6]
Effective dimension of vacuum blocks	finite product bound	RCA_0	enters explicit certification budget [7]
Thermodynamic limit, uniform (Fekete)	none in general	$\equiv \text{ACA}_0$	—
Infrared limit of c -functions, uniform	none in general	$\equiv \text{ACA}_0$	—
Cantor scaling exponent, uniform; $D = \frac{1}{4} + \frac{1}{2}\Omega$	none	$\equiv \text{ACA}_0$; not RCA_0	not certifiable [4]
Connectivity set C^X ; Ω -comb	none	$\equiv \text{ACA}_0$; not RCA_0	undecidable, no critical scale [5]
Finite-dimensional game values $G \mapsto \omega^*(G)$	none ($\text{MIP}^* = \text{RE}$)	ACA_0 ; not RCA_0	bounds not all provable [6]
Resistance exponent ζ (Lypunov)	unknown	open	estimated numerically [8]
Injective $\Leftrightarrow$ semidiscrete (Connes)	—	open	used in [7]

For the game values, $\omega^*(G)$ denotes the tensor-product value, the supremum over finite-dimensional strategies. Its lower cut $\{(G, q) : \omega^*(G) > q\}$ is Σ_1^0 , by search over finite-dimensional strategies with rational entries, so the function $G \mapsto \omega^*(G)$ exists in ACA_0 . By $\text{MIP}^* = \text{RE}$ [19] it is not computable, so it does not exist in REC . The commuting-operator value $\omega_{\text{co}}(G)$ is approached from above by the NPA hierarchy of semidefinite relaxations [17]. Hence its upper cut $\{(G, q) : \omega_{\text{co}}(G) < q\}$ is Σ_1^0 . The difference $\omega_{\text{co}}(G) - \omega^*(G)$ can be positive, and this is the Tsirelson gap. Relativistic quantum field theory keeps local measurements in spacelike separated regions out of this gap [7].

The table shows the pattern stated in the introduction. Every limit that entered an explicit certification budget has an explicit modulus and is provable in RCA_0 . Every limit that was shown to be empirically inaccessible is defined without a computable modulus, and its existence needs ACA_0 . This is not a coincidence. Proposition 4.1 links logical strength to the existence of a computable modulus. A computable modulus is exactly what a finite procedure needs in order to know when it has reached a given precision.

6 Discussion

Feferman’s thesis is that scientifically applicable mathematics needs only a system conservative over Peano arithmetic [3]. Our results refine it for physical limits. The uniform existence theorems for thermodynamic limits, infrared limits and fractal exponents are equivalent to ACA_0 . This is strictly stronger than RCA_0 , but still conservative over PA . Individual physical quantities with explicit error bounds need only RCA_0 . The infinite in physics is therefore real in a precise

sense: the general limit theorems cannot be proved from computable set existence alone. Yet it is logically mild, since it adds no new arithmetical consequences.

Two items remain open. The resistance exponent of the critical diamond-lattice cluster is a Lyapunov exponent of a random nonlinear recursion. We estimated it numerically, but we have neither a proof that the defining limit exists nor a closed form or computable modulus. Ergodic theorems are a natural tool here, and the pointwise ergodic theorem is equivalent to ACA_0 over RCA_0 [18]. An explicit contraction estimate for the recursion would instead give a computable modulus, and the existence would then be provable in RCA_0 by Proposition 4.1. The classification of injective von Neumann algebras, used in [7], rests on Tomita–Takesaki theory and ultrapower techniques. Separable Banach spaces can be coded in RCA_0 [2], but we know of no calibration of the theory of von Neumann algebras in second-order arithmetic, and the strength of Connes’ theorem is open.

Together with the earlier papers, the picture is consistent. Physical predictions for finite experiments are computable. Explicit finite-size and resource bounds are provable from computable set existence. The genuinely infinite content, including uncomputable dimensions, limit connectivity and the values of infinite-dimensional quantum games, sits exactly where arithmetical comprehension is needed and where finite experiments cannot reach.

Data and code availability

The script `reverse_math_moduli.py`, the data (CSV), the figure and the console output are available at <https://github.com/Ruqing1963/reverse-math-physics> and archived under DOI 10.5281/zenodo.23212848.

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